

Song-Ze (Jimmy) Yu

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Research Interests

My research focuses on grounded **music understanding** in large audio-language models and **controllability for musicians** in music and audio effects generation. My long-term goal is to bring music models toward **artificial music intelligence**, offering musicians new interfaces to directly discuss, collaborate, and interact with machines.

Education

University of California, Berkeley Aug. 2025 - May 2026

Visiting Student in Computer Science

- GPA: 3.9 / 4.0
- Selected Coursework: Machine Learning, LLM Post-Training & AI Agents, Deep Reinforcement Learning, Computer Vision, Computer Programming for Music Applications (CS 189, CDSS 94, CS 185, CS 180, MUSIC 159).

National Tsing Hua University Sep. 2023 - Jun. 2026

B.S. in Computer Science

- GPA: 3.91 / 4.30

National Cheng Kung University Sep. 2022 - Jun. 2023

B.S. in Computer Science and Information Engineering

- GPA: 4.16 / 4.30; Rank: **5 / 113** (Top 5%)

Publications (* indicates equal contribution)

Music Understanding in Audio-Language Models

- [1] [PitchBench: Measuring Pitch Hearing in Audio-Language Models](#) **NeurIPS 2026**
Milan Liessens Dujardin*, **Song-Ze Yu***, Craver Corby Thomas-Smith, David M. Chan, Karina Nguyen
Evaluations & Datasets
- [2] UniFront: Revisiting Time-Frequency Frontends for Unified Music and Speech Modeling
Song-Ze Yu, David M. Chan
- [3] Diagnosing Music Understanding in Audio-Language Models: Encoder-Decoder Probing and Task Adaptation
Jun-You Wang, **Song-Ze Yu**, Xuanjun Chen, Chih-Kai Yang, Ke-Han Lu, Sung-Feng Huang
- [4] [MuSP-Bench: Advanced Multimodal Benchmarking of Music Understanding across Score and Performance](#) **ISMIR 2026**
Milan Liessens Dujardin, **Song-Ze Yu**, Kevin Miao
Late-Breaking Demo
- [5] [Unifying Score and Performance for Fine-Grained Music Understanding in Audio-Language Models](#)
Milan Liessens Dujardin, **Song-Ze Yu**, Kevin Miao

Controllability for musicians

- [6] [InstructFX2FX: A Multi-Turn Text-to-Effect System for Sequential Audio Effect Refinement](#) **DAFx 2026**
Song-Ze Yu, Milan Liessens Dujardin, Yuxuan Cai, Wantong Zhang, Brian Cruz, Jeremy Wagner, Carmine-Emanuele Cella
Demonstration
- [7] MIDI-REPA: Improving Text-to-Music Generation with Symbolic Representation Alignment
Song-Ze Yu, Fang-Duo Tsai, Yi-Hsuan Yang

HCI & human simulation

- [8] [Anamnesis: An Open-Source Platform for Large-Scale Backstory-Conditioned Survey Simulation](#) **EMNLP 2026**
Song-Ze Yu, Joseph Suh, Serina Chang, David M. Chan
System Demonstration

Research Experience

Music and AI Lab, National Taiwan University Jun. 2026 - Present

Research Assistant

Advised by Prof. Yi-Hsuan Yang

- Leading the development of a **large audio-language model (LALM) for music**, bringing together insights from prior music benchmarking [1,4], methodology [2,5], and diagnostic studies [3] into large-scale model alignment and training, with collaborators from NVIDIA, BAIR, NTNU, Queen Mary University of London, and Prof. Hung-yi Lee's group.
- Researching symbolic representation alignment for text-to-music generation and representation transfer and utilization in audio-language models, leading to **MIDI-REPA** [7] and **encoder-decoder probing studies** [3].

Berkeley AI Research (BAIR) Lab, UC Berkeley Feb. 2026 - Present

Researcher

Advised by David M. Chan and Prof. Trevor Darrell

- Revisiting the ubiquitous 25 ms log-Mel frontend used across general audio encoders and models under conflicting speech-music resolution demands, leading to **UniFront** [2].

- Lead-developed **PitchBench** [1] and **Anamnesis** [8], spanning the basics of music understanding in audio-language models and large-scale human simulation.
- Center for New Music and Audio Technologies Lab, UC Berkeley**
Undergraduate Research Lead

Sep. 2025 - Jun. 2026
Advised by Prof. Carmine-Emanuele Cella
- Led the research direction on interactive, musician-centered audio effect modeling and editing, extending single-shot methods toward iterative refinement and leading to **InstructFX2FX** [6].
- Acoustics and Hearing Group, National Tsing Hua University**
Undergraduate Researcher

Jul. 2024 - May 2025
Advised by Prof. Yi-Wen Liu
- Developed the **audio-to-equalizer model (VTR)** for single-shot tone replication as my undergraduate capstone project, later deployed as a Virtual Studio Technology (VST) plugin at Positive Grid.

Selected Engineering Experience

- BerkeleyTime**
Machine Learning Pod — *Pod Lead*

Berkeley, CA
Jan. 2026 - May 2026
- Led a **10-member machine learning team** across semantic search, recommendation system, and enrollment prediction, coordinating researcher-engineer pairs and designers from model development through deployment.
 - Led development of a **new Explore page**, integrating course recommendations into BerkeleyTime’s product experience.
- Catalog Pod** — *Machine Learning Engineer*

Sep. 2025 - Dec. 2025
- Built **BGE text-embedding and FAISS vector-search services** with FastAPI for **122K+ monthly users**.
 - Reduced GraphQL first-load latency from **25s to 1s** with pagination and Redis caching; shipped and fixed **20+ pull requests**.
- Positive Grid**
Machine Learning Intern

Jun. 2025 - Aug. 2025
Taipei, Taiwan
- Served as Scrum Master for **Amp-AI-SaaS** and designed the **Multi-Agent Orchestration Harness** for ongoing software-engineering workflows at Positive Grid.
 - Integrated my capstone **VTR audio-to-equalizer model** into a functional **JUCE VST plugin**, integrating PyTorch inference for real-time audio processing in a digital audio workstation.

Awards & Leadership

- Meichu Hackathon — 3rd Place, NXP Group**

Sep. 2026
- Built **GodEyes**, a **physical AI** system using a **world model** to turn crime-scene photos into explorable 3D worlds.
- Claude × a16z × Berkeley M.E.T. Makeathon — Best Design**

Oct. 2025
- Built Haven in five hours, a mobile app that personalizes home-design recommendations from room scans.
- Meichu Hackathon — 2nd Place, Google Group**

Oct. 2024
- Built **Travelity**, an AI travel planner using interests, personality, and real-time map and search data.
- Taipei Hackathon — 3rd Place**

Sep. 2024
- Built **ParkFlow**, a WebView microservice for real-time parking information, navigation, and notifications; selected for integration into Taipei’s official city app (**3M+ downloads**) and awarded NT\$100K.
 - Produced a **hackathon vlog** (**1K+ views**); **invited by the Taipei City Government** to share the experience a year later.
- SITCON Hackathon — Group Award Winner**

Jul. 2024
- Built **ccDiary**, an AI-assisted mental-health journal with emotion analysis and anonymous social features.
- National Cheng Kung University — Campus Speaker**

Aug. 2023
- Delivered an **invited talk** for incoming students on writing music as a diary, then turned audience-submitted phrases and sentences on Slido into improvised songs performed live.

Selected Music Experience

- Pianist, Composer & Producer**

Music CV | YouTube (7K subscribers)
- Performed **24 solo piano recitals** across Greater China and appeared on **70+ TV shows and New Year galas** (春晚) between 2009 and 2016.
 - Shared the stage with **Jay Chou** (周杰倫) **three times** and performed eight-hands piano at Taipei Arena with Bai-An (白安) and Cosmos People (宇宙人) as a guest at Yan Jue’s (嚴爵) concert.
 - Wrote **20+ original songs with 20K+ views** and continues to compose, perform, and produce original music.